

contains source and destination addresses and ports. As mentioned in the previous articles of this series, we are aware that the TCP/IP communication comprises a three-way handshake (SYN, SYN-ACK, ACK), which sets up a communication channel between two network interface cards. The data flows between them through this channel, and it is terminated by using a FIN/FIN-ACK handshake.

Earlier articles also covered the fact that each of these packet fields can be modified by hackers for their benefit. The source and destination IP address or ports are very commonly attacked fields in the denial of service and spoofing attacks, as well as in network penetrations. Here, it is important to understand that it's not just these fields that could be modified, but in fact, each and every flag and field in a TCP frame and the underlying Ethernet frame can be modified or altered—all for the purpose of network penetration. Please note that packet crafting and packet spoofing are often mistakenly assumed to be the same technique; however, they are very different from the impact standpoint. Spoofing is used by hackers to hide their identities and presence on the network. Spoofing is primarily used to gain network information such as open ports, running services, active hosts, etc, during which time the target host fails to trace the attacker. On the other hand, packet crafting takes a step further by trying to test the presence, functionality or the accuracy of target network's firewall rules, and intrusion detection systems. Packet crafting requires in-depth knowledge of TCP packets and how they work, and is more of a manually orchestrated attack than a programmatic one. This makes it a technically advanced way of trying to hack into networks.

Packet crafting in action

Packet crafting is a task that is methodically carried out to penetrate into a network's infrastructure. Please refer to Figure 2, which shows the four distinct steps involved in the packet crafting attack. Let's understand each of these steps a bit more in detail.

Packet assembly: This is the first step in the crafting process, wherein a hacker decides which network needs to be hacked, tries to gather possible vulnerability information and creates or fabricates the packets to be sent. This packet is then checked for accuracy, especially to ensure that the attack is as 'invisible' on the network as possible, to go undetected. For example, the packet being created can have a spoofed source address and a dummy TCP sequence number. The assembly of a packet need not be done from scratch; a packet going over the wire can be captured and its contents can be modified to serve the hacking purpose.

Packet editing: In this step, usually a dry run on the



Figure 1: A basic Ethernet and TCP/IP packet

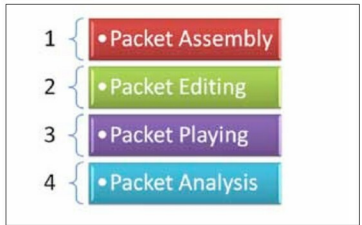


Figure 2: The steps involved in packet crafting

assembled packet is tested and based on the results gathered, and the packet is tuned up or corrected before moving to the next step. In the editing phase, the focus is usually to gather the maximum amount of information by injecting the minimum number of packets into the network. For example, to test how a firewall responds to malformed packets, a simple packet with a false source IP address and with ACK field bit set can be created. In ideal situations, the firewall should drop such a packet.

Packet playing: Once the correct packet or a stream of packets is created, 'packet playing' sends it onto the network, and collects the resultant packets to perform further analysis and co-relation. This is when an actual attack is performed. If the expected outcome is not achieved, hackers go back to the editing phase to change the attack scenario.

Packet analysing: In this process, the packet streams are gathered to decode the presented response by the target network. Hackers may use simple packet sniffing tools for this purpose, or can capture the packet streams in the form of a log file and analyse it. This step either provides evidence to the hackers that they were successful in penetration, or at least gives